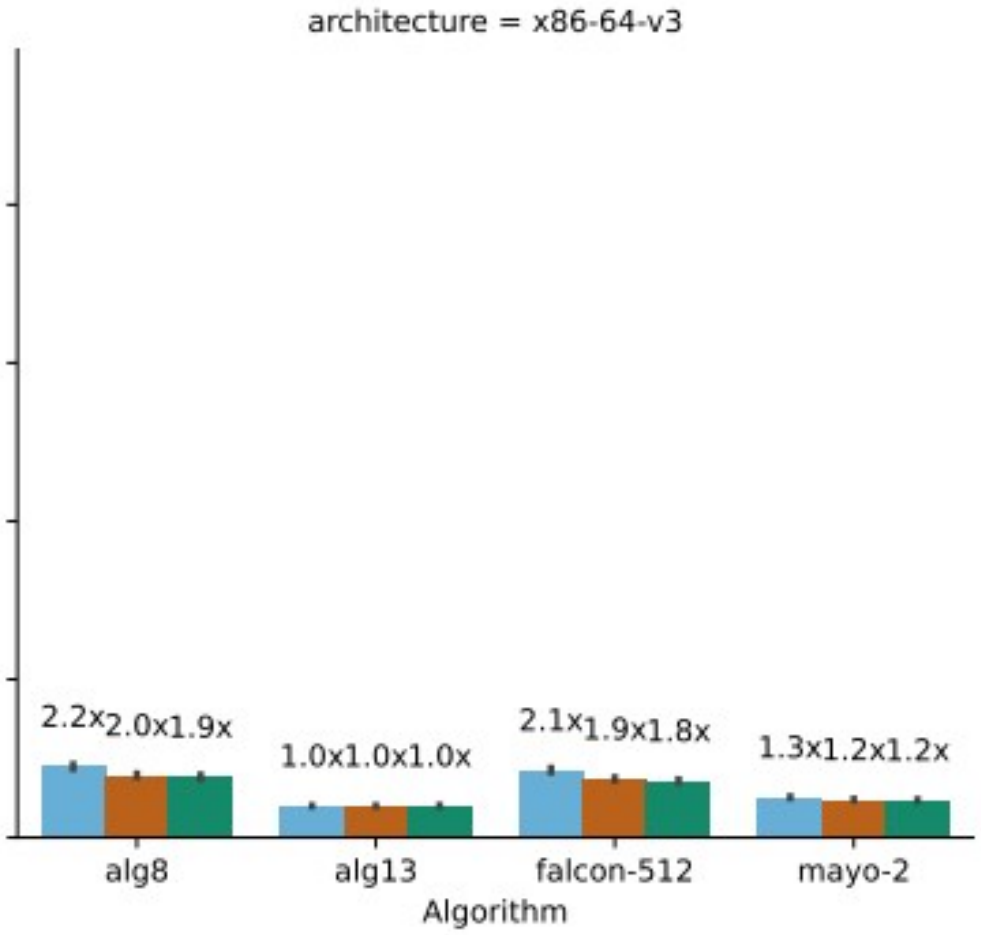
Simulating DNSSEC on a testbed based on real resolver data

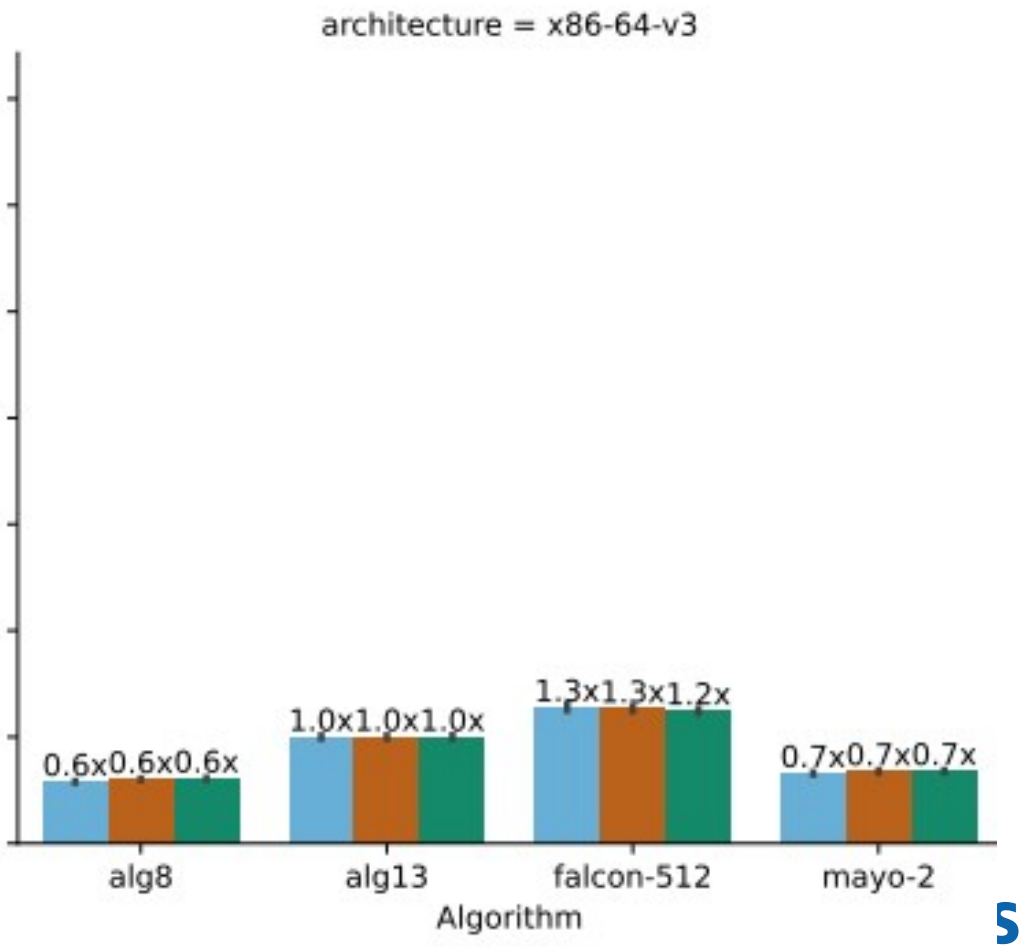
Elmer Lastdrager – SIDN Labs

IETF 123 flashback: TLD operator POV

Signing zone file (ratio)



Validating zone file (ratio)



DNS Resolver

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SURF

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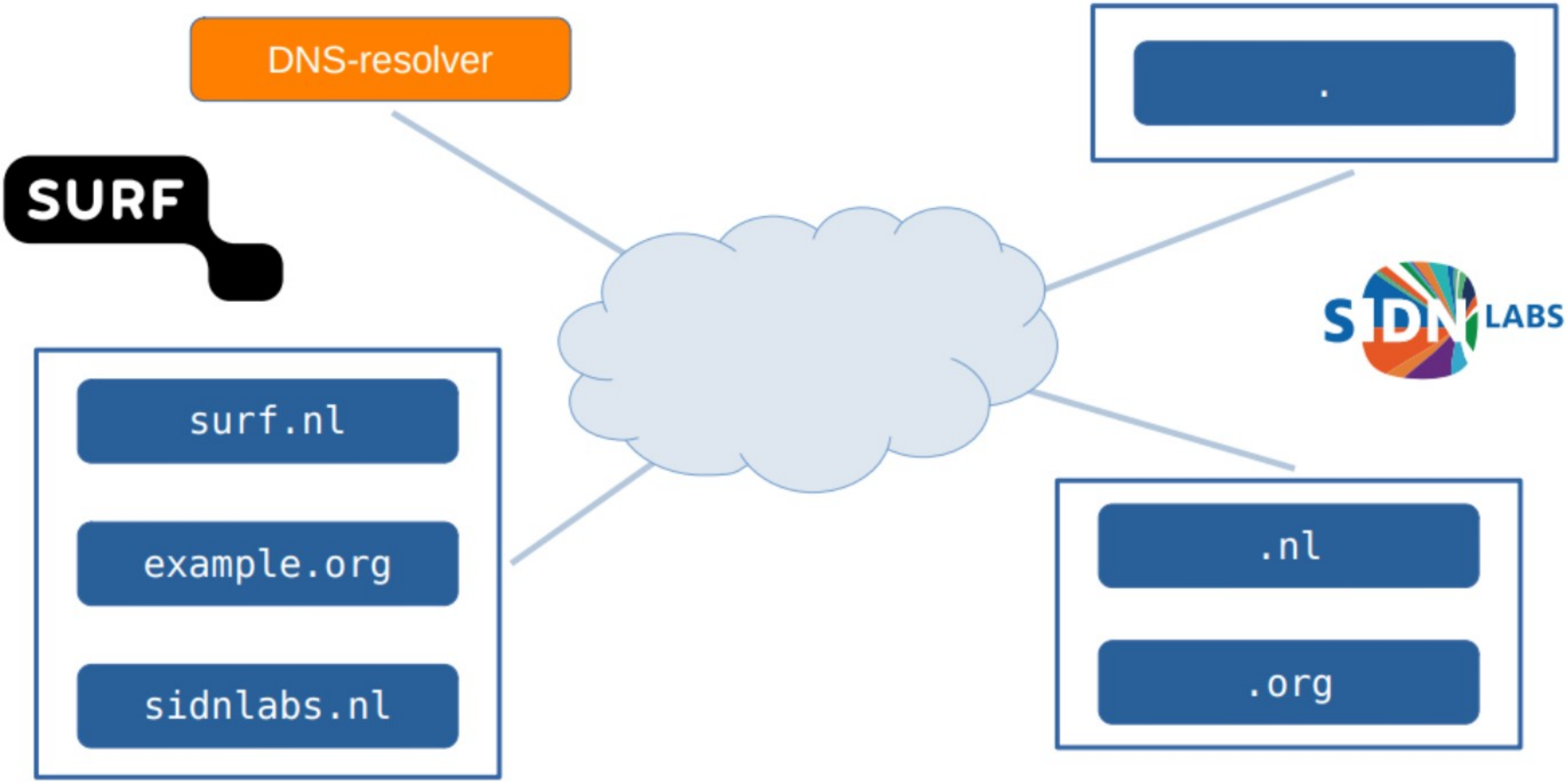

DNS-resolver
SURF
.
SIDN LABS
surf.nl
example.org
sidnlabs.nl
.nl
.org

What we do

Step 1: obtain resolver data (dnstap).

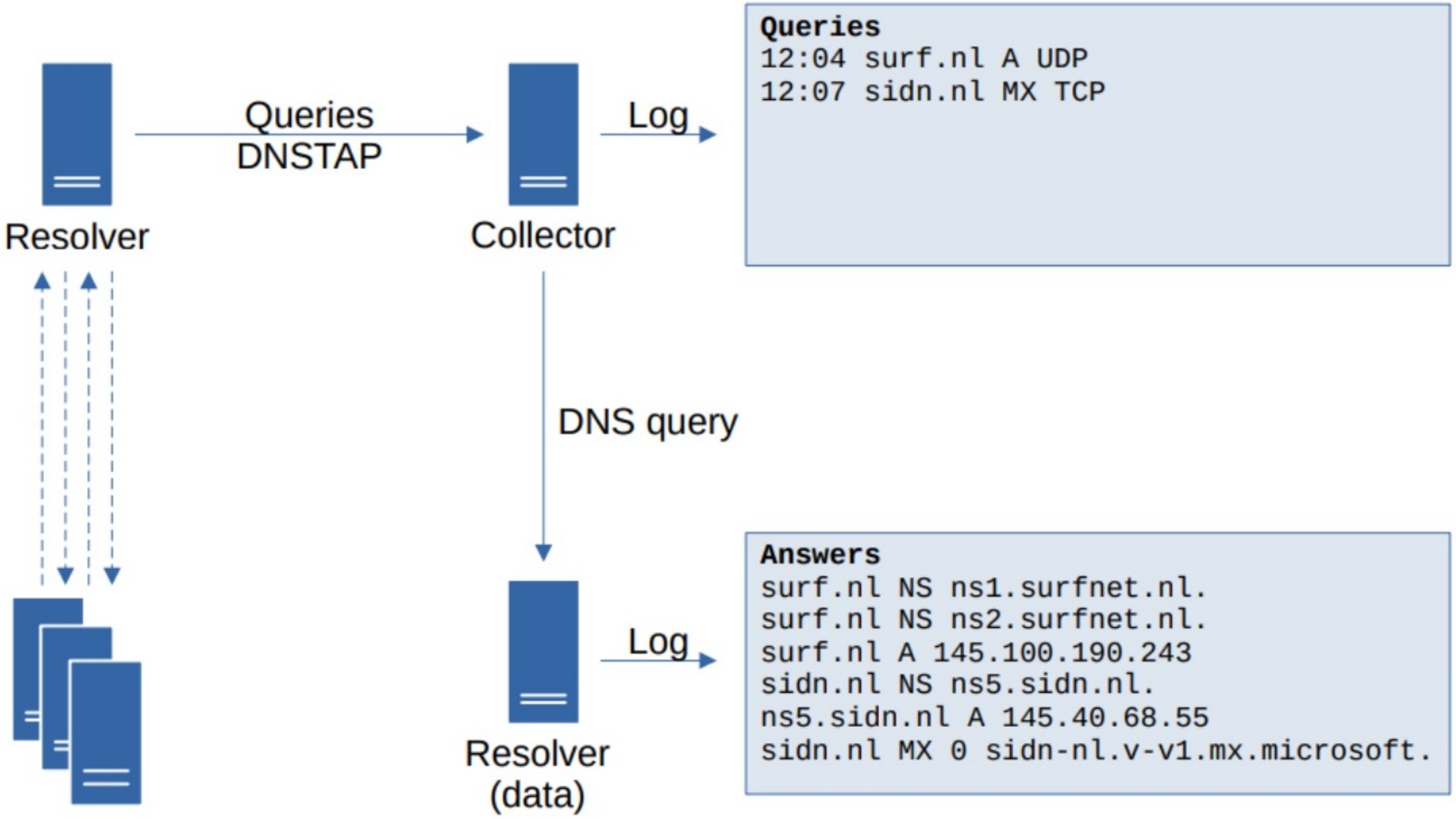
Step 2: build zones based on the answers.

Step 3: sign zones using PQC algorithms.

Step 4: replay queries.

Notes on privacy

1. We do not store client IP addresses
2. Exact content of DNS message not relevant
3. But the size and number of queries are relevant
4. So, we anonymize data (*keyed transformation*?)


Queries
DNSTAP
Log
Resolver
Collector
Queries
12:04 surf.nl A UDP
12:07 sidn.nl MX TCP
DNS query
Log
Resolver
(data)
Answers
surf.nl NS ns1.surfnet.nl.
surf.nl NS ns2.surfnet.nl.
surf.nl A 145.100.190.243
sidn.nl NS ns5.sidn.nl.
ns5.sidn.nl A 145.40.68.55
sidn.nl MX 0 sidn-nl.v-v1.mx.microsoft.

```
Answers
surf.nl NS ns1.surfnet.nl.
surf.nl NS ns2.surfnet.nl.
surf.nl A 145.100.190.243
sidn.nl NS ns5.sidn.nl.
ns5.sidn.nl A 145.40.68.55
sidn.nl MX 0 sidn-nl.v-v1.mx.microsoft.
```

```
Answers
r5nr.nl NS mq8.ql9c1h8.nl.
r5nr.nl NS loq.ql9c1h8.nl.
r5nr.nl A 48.194.99.216
4itd.nl NS ecr.4itd.nl.
ecr.4itd.nl A 166.246.223.124
4itd.nl MX 5 r9p1jlg.8xrl.ym.1qz1utjhe.
```

```
$ORIGIN r5nr.nl.
r5nr.nl. 3600 SOA ...
@ NS mq8.ql9c1h8.nl
@ NS loq.ql9c1h8.nl
@ A 48.194.99.216
```

```
$ORIGIN 4itd.nl.
4itd.nl. 3600 SOA ...
@ NS ecr.4itd.nl.
ecr A 166.246.223.124
@ MX 5 r9p1jlg.8xrl.ym.1qz1utjhe.
```

podman

TCP

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M 1
90
1.25

Up next

Run and analyze algorithms:
Falcon, Mayo, SQISign,
RSA (8), ECC (13)

Adding SNOVA (Radboud University)

MTL containers from Verisign (recent)

Go beyond initial measurements,
automate.

Thanks for your attention

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